

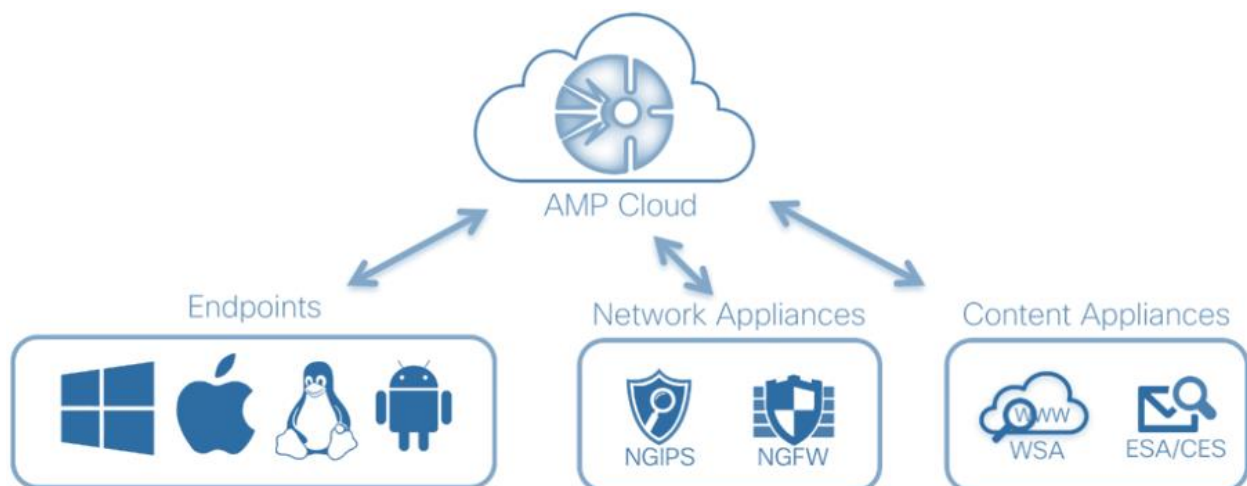
AMP (Advanced Malware Protection):

Cisco AMP for Endpoints provides next-generation endpoint protection, scanning files using a variety of antimalware technologies, including the Cisco antivirus engine.

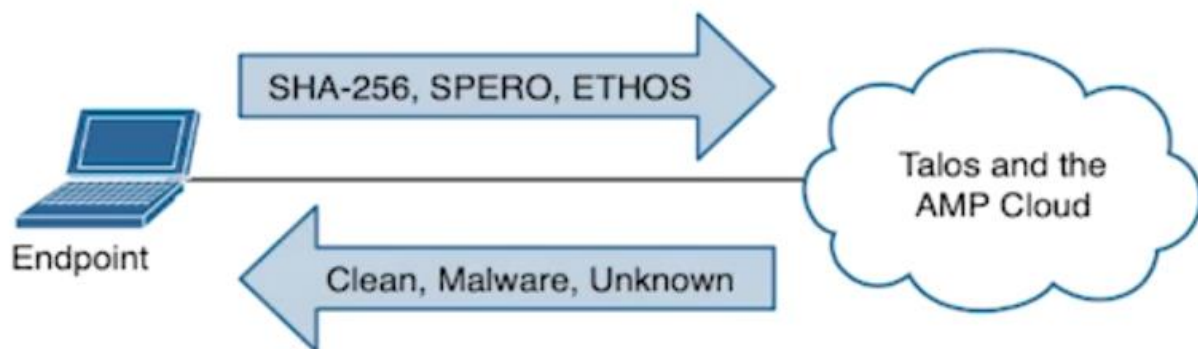
Cisco Advanced Malware Protection then goes a step further than most malware detection tools, continuously monitoring every file in your network. If Cisco AMP detects malicious behavior, it helps you remediate quickly.

The Cisco Advanced Malware Protection (AMP) is a supplementary malware defense system that protects your entire network against highly sophisticated and advanced malware attacks. Being a supplemental solution, the AMP is designed to work with other Cisco network security appliances, particularly the Next-Generation Intrusion Prevention System (NGIPS).

Cisco AMP or Advanced Malware Protection is Cisco's answer to the Next Generation of visibility, control, and protection against advanced threats for today's internet connected world. AMP gives you real-time blocking of malware and advanced sandboxing, that is backed up by world class global threat intelligence, to provide rapid detection, containment and removal of advanced malware.



AMP for Endpoints has Endpoint Connectors, which are agents on the endpoints, providing security against malware and intrusion detection. It also provides intrusion prevention. We install the Connector on all the endpoints before they're deployed and also on our virtual desktop images. They provide constant monitoring and alerting on any events or potential threats to let us know when there is something going on that we can further investigate. AMP intersects with a bunch of other Cisco tools, such as Threat Grid, Threat Response, and Talos Intelligence to identify threats, then automatically quarantine or remove them. It also gives you the ability to isolate endpoints to prevent further spread of any sort of malware, like a virus that might infect other machines.



Retrospective Security:

Retrospective security is an advanced security measure that allows organizations to prevent attacks before they happen. The AMP continuously tracks and analyzes files and file activities across your systems and compares these events to what preceded or happened in past attacks. If a file exhibits malicious behavior, AMP provides you with a retrospective alert which enables you to stop a potential threat from succeeding.

File Reputation:

Cisco AMP combines use of advanced file behavior patterns, along with collective intelligence to determine whether a file is good (safe) or bad (malicious). This allows AMP to perform more accurate detections during file analysis and inspection. File reputation is based upon a variety of factors determined over a period of time, it supports most file types, and identifies each file by its content, so is not dependent on the file extension to determine type.

Trajectory:

AMP will also track the lifecycle of any identified malware on an endpoint. In an easy timeline, the administrator can see where the attack originated, what files it influenced and when, and any other malicious files it may have downloaded to the endpoint.

Indications of Compromise (IOCs):

IOCs (also known as Indications or Indicators of Compromise) are pieces of information that can help identify specifics around abnormal or malicious behavior on a network or system. With AMP, File activity and recorded events are linked together and prioritized as potential active breaches. AMP automatically captures and links this security event data from multiple sources (such as intrusion and malware events) to help security teams connect the individual events to larger, coordinated attacks and prioritize them as high-risk. This component of the dashboard will reveal to the administrator evidence of possible attacks before entering into your network.

Antivirus Engine:

The built-in AV engines perform offline system-based detections, which includes scanning for the presence of rootkits, to complement Cisco's advanced endpoint protection capabilities such as local IOC scanning, and device and network flow monitoring. If a customer wishes to consolidate both their antivirus and their advanced endpoint protection to run under a single agent, then the AV engine can be enabled in policy to achieve this. Two engines exist in the AMP for Endpoints product; the ClamAV engine is available for running on Linux and Mac based devices, while the Tetra engine is available for Windows devices.

Static and Dynamic Malware Analysis:

A safe and highly secure sandboxing environment helps you execute, analyses, and test both malware & files with unknown reputation or behavior, in order to discover previously unknown zero-day threats. Cisco has integrated Threat Grid's advanced sandboxing, using its static and dynamic malware analysis technology, into their AMP solutions. This integration results in a more comprehensive analysis by performing checks against a large set, of more than a 1000 highly accurate behavioral indicators. Threat Grids analysis produces very few false positive results (a result which indicates something is bad when it is not), increasing your confidence, so you can make fast and accurate decisions.

Retrospective Security:

Instead of just analyzing network traffic or file when it first arrives, Cisco AMP uses continuous analysis and retrospective security, even after the initial inspection occurs. Through continuous analysis techniques, AMP monitors and records file activity and network communications, to detect if behavior changes, and help identify stealthy or malicious threats. Alerts are sent at the first sign of suspicious activity, i.e. when a file disposition changes after extended analysis (a good file turns bad); giving the security team awareness of and visibility into the malware that evades initial defenses. As AMP records all activity, it knows where these files or network communications have been seen across your organization, then using retrospective security can quickly respond and remove, or quarantine, the malware for all users and endpoints, eliminating the threat.

Prevalence:

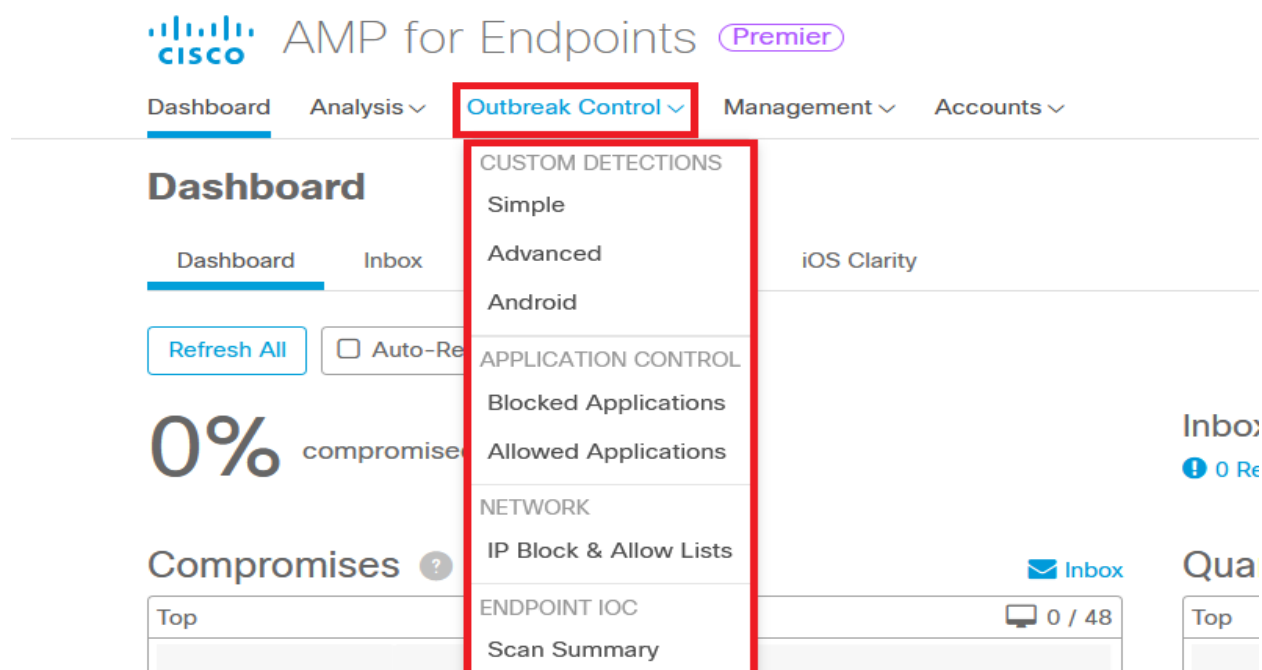
The Prevalence feature displays all files that have been run in your organization, ordered by a prevalence rating from lowest to highest. This helps become aware of previously undetected threats that have been seen by a small number of users. Generally, files that are run by a large number of users (high prevalence) tend to be valid applications. Whereas files that are run by only a few users (low prevalence) may be considered malicious (such as a targeted advanced threats) or questionable applications you may not want on your extended network. Prevalence is another powerful feature of AMP for detection of advanced threats, with all low prevalence files being automatically sent to Threat Grid for analysis.

File Trajectory:

This feature is used to continuously track files presence throughout your network environment over time, to achieve visibility and reduce the time required to scope a malware breach. File trajectory maps the file to which endpoints it has been seen on, and how the file has been transferred across network. It contains files disposition (good, bad or unknown), determines the first endpoint that saw the threat, and whether you have other hosts containing the file that are also at risk. This is a key component of AMP's retrospective security.

Outbreak Control:

Outbreak Control is how Cisco AMP actually eliminates attacks on your endpoints. It can be set to automatically remove and block the malware from spreading within or to other endpoints, or an administrator can elect to do this manually. By simply right clicking the alert, administrators can whitelist or blacklist a file.



Continuous Monitoring and Analysis:

Once a file lands on the endpoint, AMP for Endpoints continues to watch, analyze, and record all file activity, regardless of the file's disposition. If malicious behavior is detected at some point in the future, AMP can automatically block the file across all endpoints, and show the security team the entire recorded history of the malware's behavior. You can see where it came from, where it's been, and what it's doing across all of your endpoints: PC, Mac, Linux, mobile devices. This helps you understand the full scope the compromise and quickly respond.

Malware Blocking:

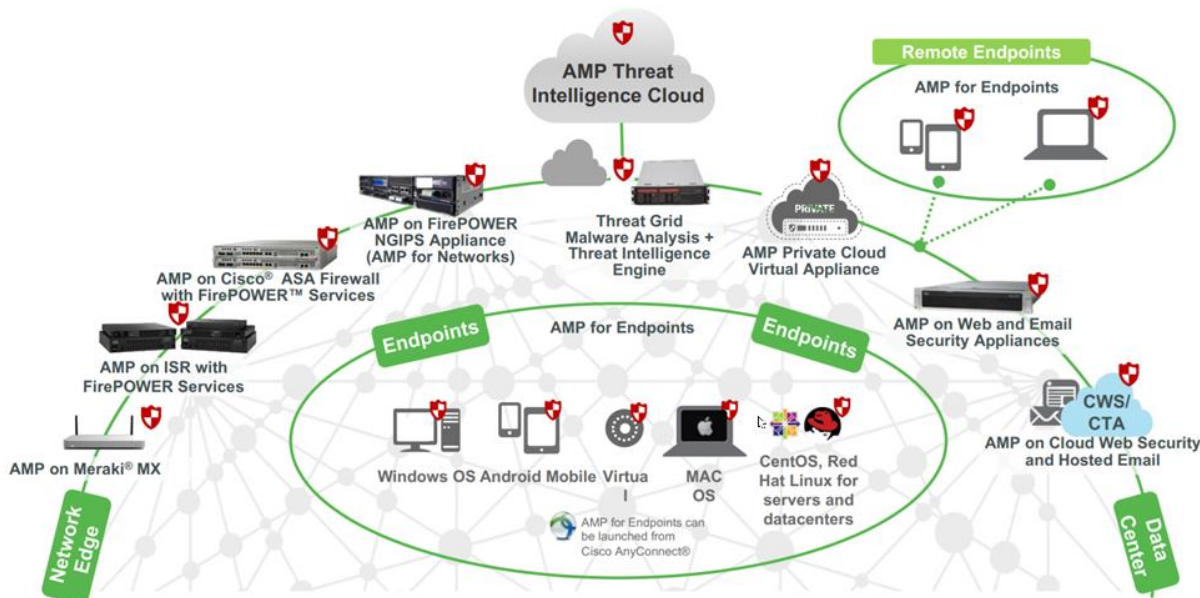
AMP for Endpoints uses a framework of complementary detection engines, including one-to-one signatures, fuzzy fingerprinting, machine learning, and an AV detection engine—all working together to catch and block malware before it can execute.

File Sandboxing:

A built-in sandbox automatically analyzes unknown files against over 700 behavioral indicators to detect malicious files and automatically block and quarantine them.

The Cloud:

Cisco AMP for Endpoints is cleverly managed through the cloud. Installing this level of analysis and security depth management directly onto a device would grind performance to a halt. Through the cloud, however, the endpoint can achieve an exceptional level of security without the burden of large security applications.



The visibility and insight this solution gives you into threats is pretty granular. It has constant monitoring. You can get onto the device trajectory to look at a threat, but you can also see what happened prior to the threat. You can see what happened after the threat. You can see what other applications were incorporated into the execution of the threat. For example, you have the event, but you see that the event was launched by Google Chrome, which was launched by something else. Then, after the event, something else was launched by whatever the threat was. Therefore, it gives you great detail, a timeline, and continuity of events leading up to whatever the incident is, and then, after. This helps you understand and nail down what the threat is and how to fix it.

